PRAGATI ENGINEERING COLLEGE (A)

SENTINELVISION-AI

Smart Computer Vision & Attentiveness Monitoring System

Project Documentation:

Submitted by:

L. Bhargav - 25A31A05FQ

V. Shaun Ajit - 25A31A05GF

Satyaveni - 25A31A05EA

M. Maitri - 25A31A05EK

Department:

Computer Science and Engineering (CSE)

Academic Year: 2026-27

Live Prototype: https://sentinelvisionlive.netlify.app

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

1. Abstract :

SentinelVision AI is a web-based real-time monitoring system that uses a laptop's built-in webcam or an external webcam to observe whether a person is attentive and present. It detects three abnormal conditions: eyes remaining closed for longer than a set duration, the face being hidden or covered, and the person being absent from the camera frame. Natural eye blinking is recognised and ignored, so it does not cause false alarms.

When an abnormal condition continues beyond its configured limit, the system raises an audible siren and records the event in an incident history. A dark, security-style dashboard shows the live camera feed, system status, eye and face states, the Eye Aspect Ratio (EAR) value, alert counters, and a searchable incident log that can be exported as CSV.

All video is processed locally on the user's device. No webcam footage is uploaded or stored on any server. The system uses JavaScript, browser-based computer vision, and the MediaDevices API, and works with built-in cameras as well as virtual webcams such as Iriun Webcam.

The main objective of the project is to provide a low-cost, privacy-friendly attentiveness monitor that can help in situations such as driver alertness, study sessions, remote exam supervision, and workplace safety.

KEY POINTS

- Detects closed eyes, hidden face and person absence in real time.
- Ignores natural blinks, so there are no false alarms.
- 100% on-device processing: no video is uploaded or stored.

2. Introduction :

Loss of attention is a common cause of accidents and reduced productivity. A driver who closes their eyes for a few seconds, a student who dozes off while studying, or an operator who leaves a workstation unattended can all create risk. Dedicated monitoring hardware is often expensive, and many camera-based tools send video to cloud servers, which raises privacy concerns.

Modern browsers can access the camera and run computer vision models directly on the user's device. SentinelVision AI uses this capability to build a monitoring tool that needs only a webcam and a web browser.

The system tracks facial landmarks around the eyes, calculates the Eye Aspect Ratio (EAR) for every frame, and compares the result with a configurable threshold. A timer-based check then separates a short, natural blink from prolonged eye closure. Similar timers track how long the face has been hidden and how long the person has been absent.

The project is delivered as a web application with a professional dashboard, so that the monitoring process, alerts, and incident records are easy to follow.

KEY POINTS

- Needs only a webcam and a modern web browser.
- Eye Aspect Ratio (EAR) plus timers separate blinks from real alerts.
- A live dashboard shows status, counters and incident history.

3. Problem Statement :

In many situations, a person's attentiveness has to be checked manually or not at all. Existing solutions have several limitations:

- Continuous human supervision is tiring and impractical.
- Specialised drowsiness-detection hardware is costly and not widely available.
- Many camera-based tools upload video to external servers, creating privacy risks.
- Simple detectors treat every blink as an alert, producing frequent false alarms.
- Most tools do not keep a record of incidents with their time, type, and duration.
- Thresholds are often fixed and cannot be adjusted to different users or lighting conditions.

Therefore, a simple, configurable, browser-based system is needed that can detect prolonged eye closure, hidden faces, and person absence in real time, raise an audible alarm, keep an incident log, and process all video locally.

KEY POINTS

- Need: low-cost, configurable, private and real-time monitoring.
- Existing tools cause false alarms or upload video to servers.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

4. Objectives of the Project :

The main objective of SentinelVision AI is to develop a real-time webcam-based monitoring system that detects abnormal attentiveness conditions and alerts the user immediately.

The specific objectives are:

1. To access the laptop or external webcam securely using the browser's MediaDevices API.
2. To track facial landmarks and calculate the Eye Aspect Ratio (EAR) in real time.
3. To distinguish natural blinks from prolonged eye closure.
4. To detect when the face is hidden, covered, or not visible.
5. To detect when the monitored person is absent from the frame.
6. To trigger an audible siren with different messages for different incidents.
7. To provide adjustable thresholds, alarm volume, and alarm options through a settings panel.
8. To record every incident automatically with its number, timestamp, type, severity, duration, and description.
9. To allow incident history to be exported as a CSV file or cleared.
10. To process all video locally and protect the user's privacy.
11. To present everything through a modern, responsive, dark-themed dashboard.

KEY POINTS

- Goal: accurate detection, instant audible alerts and full privacy.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

5. Technologies Used :

The SentinelVision AI project uses the following technologies.

5.1 HTML

HTML builds the structure of the dashboard, including the video panel, status indicators, counters, settings panel, and incident history table.

5.2 CSS

CSS provides the futuristic cybersecurity-inspired design: dark navy and black backgrounds, cyan and electric blue highlights, red alert states, green normal states, smooth animations, and a responsive layout for desktop and mobile screens.

5.3 JavaScript

JavaScript is the core language of the project. It controls camera access, the detection loop, state tracking, alarm control, settings storage, incident recording, and CSV export.

5.4 MediaDevices API (getUserMedia)

The browser's MediaDevices API is used to list available cameras, request camera permission, and stream live video into the dashboard. It also supports virtual cameras such as Iriun Webcam.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

5.5 Browser-Based Computer Vision Library

A browser-compatible facial landmark library (such as MediaPipe Face Mesh) is used to locate the face and eye landmarks on each frame. All processing runs locally in the browser.

5.6 Web Audio API

The Web Audio API generates the siren sound and controls its volume, so no external audio files or servers are needed.

5.7 Browser Local Storage

Local storage keeps the user's saved settings between sessions on the same device.

5.8 Netlify

Netlify is used to host the live prototype at https://sentinelvisionlive.netlify.app.

5.9 Visual Studio Code

Visual Studio Code is the development environment used to write and manage the source code.

5.10 Git and GitHub

Git is used for version control, and GitHub is used to store the source code and support collaboration among team members.

KEY POINTS

- Everything runs in the browser: HTML, CSS, JavaScript, MediaDevices API and Web Audio API.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

6. System Requirements :

6.1 Hardware Requirements

- Processor: Intel Core i3 or equivalent and above (Core i5 recommended for smooth real-time detection).
- RAM: Minimum 4 GB; 8 GB recommended.
- Storage: Minimum 1 GB of free space for project files.
- Camera: Built-in laptop webcam, external USB webcam, or a phone camera through Iriun Webcam.
- Display: Standard laptop or desktop display.
- Speakers or headphones: Required for the audible alarm.
- Internet connection: Required to open the hosted application and load the vision library.

6.2 Software Requirements

- Operating System: Windows 10 or Windows 11.
- Programming Language: JavaScript.
- Frontend: HTML and CSS.
- Web Browser: Google Chrome or Microsoft Edge (latest version) with camera permission enabled.
- Optional: Iriun Webcam app on phone and Iriun Webcam software on PC for virtual camera support.
- Code Editor: Visual Studio Code.

KEY POINTS

- Works on any Windows laptop with a webcam and Chrome or Edge.
- Phone cameras work too through Iriun Webcam.

SentinelVision AI
Smart Computer Vision for Attentiveness Monitoring

7. System Design :

SentinelVision AI follows a client-side architecture in which every stage of processing happens inside the user's browser. It consists of four main layers.

7.1 Presentation Layer

The presentation layer is the dashboard the user sees. It includes:

- Branding and system status (Standby, Monitoring, Alert).
- Monitoring timer.
- Live camera feed with camera selection and connection controls.
- Eye State, Face State, and EAR displays.
- Alarm status and alert counters.
- Settings panel and incident history table.

7.2 Capture Layer

The capture layer uses getUserMedia() to obtain the video stream. It manages camera selection, permission requests, connection status, and error handling for camera disconnection or denied permission.

7.3 Detection and Logic Layer

This layer analyses each frame and decides whether the situation is normal or abnormal. It performs:

- Face and landmark detection.

SentinelVision AI
Smart Computer Vision for Attentiveness Monitoring

- EAR calculation for both eyes.
- Blink filtering using time thresholds.
- Face-hidden and person-absent timing.
- Alert decision-making.

7.4 Response and Storage Layer

This layer reacts to the decisions of the detection layer. It plays or silences the siren, updates the alert counters, creates incident records, stores the settings, and prepares the CSV export.

Eye Aspect Ratio (EAR)

EAR is calculated from six landmark points around each eye (P1 to P6):

```
EAR = ( |P2 - P6| + |P3 - P5| ) / ( 2 x |P1 - P4| )
```

When the eye is open, EAR stays roughly constant. When the eye closes, EAR drops close to zero. If the value falls below the configured threshold (default 0.22), the eye is marked CLOSED.

KEY POINTS

- Four layers: Presentation, Capture, Detection and Logic, Response and Storage.
- Default EAR threshold is 0.22; below it the eye is CLOSED.

8. System Flowchart :

The following flow represents the basic working of SentinelVision AI.

1. START - User opens the SentinelVision AI website.
2. Connect the camera and allow browser permission.
3. Select a camera and click Start Monitoring.
4. Capture a video frame.
5. Is a face detected?
   - No: Start or continue the face-hidden / person-absent timer. If the timer exceeds its threshold, trigger the alarm and record an incident.
   - Yes: Reset the face-hidden and absence timers and continue to the next step.
6. Calculate EAR and compare it with the threshold.
   - EAR above threshold: Eyes OPEN. Reset the eyes-closed timer.
   - EAR below threshold: Eyes CLOSED. Start or continue the eyes-closed timer.
7. If the eyes reopen before the duration limit, count it as a natural blink. If the eyes stay closed beyond the limit, trigger the alarm and record an incident.
8. When normal conditions resume, silence the alarm (if Auto-Silence is enabled).
9. Repeat from step 4 until the user clicks Stop Monitoring.
10. END.

KEY POINTS

- Three checks every frame: Face detected? EAR below threshold? Time limit exceeded?
- The loop repeats until the user clicks Stop Monitoring.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

9. Module Description :

SentinelVision AI is divided into the following modules.

9.1 Dashboard Module

Displays the SentinelVision AI branding, system status, the digital monitoring timer, Eye State, Face State, EAR value, alarm status, and all alert counters. Status colours change between green (normal) and red (alert).

9.2 Camera Module

Handles camera access using the MediaDevices API.

- Lists available cameras and lets the user select one.
- Requests camera permission.
- Shows live video and the connection status.
- Supports Iriun Webcam and other virtual cameras.
- Handles disconnection and permission errors.

9.3 Eye Detection Module

Tracks facial landmarks, calculates EAR, and labels the eyes as OPEN or CLOSED. Closures shorter than the configured duration are counted as natural blinks and do not raise an alarm. Closures longer than the limit raise an Eyes Closed alert.

9.4 Face and Person Detection Module

Checks whether a face is visible. If the face disappears but the person may still be in view, it is treated as a Face Hidden condition (for example, a covered face). If no person is visible at all, it is treated as a Person Absent condition. Each condition has its own timer and threshold.

9.5 Alarm Module

Produces an audible siren using the Web Audio API.

- Adjustable volume.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

- Enable or disable audible alarms.
- Auto-silence when conditions return to normal.
- Different alert messages for eyes closed, face hidden, and person absent.
- A Test Alarm button to check the sound.

9.6 Settings Module

A settings panel allows the user to adjust:

Setting	Default Value
EAR sensitivity threshold	0.22
Eyes-closed duration threshold	1.0 s
Face-hidden duration threshold	1.0 s
Person-absence timeout	1.0 s
Alarm volume	85%
Enable audible alarms	On
Auto-silence on recovery	On

The user can save the settings or restore the defaults.

9.7 Incident History Module

Records every abnormal event automatically in a real-time table with: incident number, timestamp, incident type, severity, duration, and description. The user can export the table as a CSV file or clear the history.

KEY POINTS

- Seven modules: Dashboard, Camera, Eye Detection, Face and Person, Alarm, Settings, Incident History.

10. Implementation :

SentinelVision AI is implemented as a browser-based web application. Video frames are processed on the user's device only.

10.1 Frontend Implementation

HTML defines the dashboard layout. CSS creates the dark navy and black theme with cyan and electric blue highlights, red alert states, green normal indicators, smooth transitions, and a layout that adapts to desktop and mobile screens. JavaScript updates the live indicators, counters, timer, and incident table without reloading the page.

10.2 Camera Implementation

The application calls navigator.mediaDevices.enumerateDevices() to list cameras and navigator.mediaDevices.getUserMedia() to start the selected camera. The stream is attached to a video element. The code reports the connection status and shows a clear message if permission is denied or the camera is disconnected.

10.3 Detection Implementation

For every processed frame:

1. The face landmark model returns the landmark points of the face.
2. The six points around each eye are used to calculate the left and right EAR.
3. The average EAR is compared with the sensitivity threshold.
4. The time spent in the current state is tracked with timestamps.
5. A blink is counted when the eyes close and reopen within the allowed duration.
6. An alert is raised only when a state lasts longer than its configured threshold.

To keep detection fast, the system processes frames only while monitoring is active, avoids repeated calculations, and keeps only the most recent values it needs.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

10.4 Alert and Incident Handling

When a threshold is exceeded, the system starts the siren, shows a warning banner with a message specific to the incident, increases the matching counter, and creates an incident record. When the incident ends, the record is completed with its total duration and the siren stops if Auto-Silence is enabled.

10.5 Privacy

All processing is done locally in the browser. No video frame is uploaded, streamed to a server, or stored. A privacy notice on the dashboard states that the system is 100% on-device.

10.6 Testing

The prototype can be tested using the following operations:

1. Open the application and check that the dashboard loads correctly.
2. Click Connect and allow camera permission; confirm that live video appears.
3. Select a different camera and verify that the feed changes.
4. Click Start Monitoring and confirm the status changes to Monitoring.
5. Blink normally and check that no alarm sounds and the Natural Blinks counter increases.
6. Keep the eyes closed beyond the threshold and confirm that the alarm and an Eyes Closed incident appear.
7. Cover the face and confirm a Face Hidden alert.
8. Move out of the frame and confirm a Person Absent alert.
9. Return to normal and verify that the alarm silences automatically.
10. Change the settings, save them, and restore the defaults.
11. Export the incident history as CSV and clear the history.
12. Disconnect the camera or deny permission and verify that a clear error message is shown.

KEY POINTS

- All detection runs locally per frame; alerts fire only after the configured time limit.

SentinelVision AI
Driver Computer Vision & Attentiveness Monitoring

11. Future Scope :

SentinelVision AI can be extended with the following features.

- Yawn and Head-Pose Detection: Detect yawning and head nodding for improved drowsiness detection.
- User Accounts: Login and profiles for individual settings and history.
- Cloud Dashboard (Optional): Share incident summaries, not video, with supervisors.
- Notifications: Send email, SMS, or mobile alerts when incidents occur.
- Analytics and Reports: Charts of attentiveness over time and downloadable reports.
- Multi-Person Monitoring: Monitor more than one person in the same frame.
- Low-Light Improvement: Better performance in dim lighting.
- Mobile and Desktop Apps: Installable versions for easier daily use.
- Admin Panel: Central management of settings for institutions or organisations.

KEY POINTS

- Next steps: yawn detection, analytics, notifications and multi-person monitoring.

SentinelVision AI
Smart Computer Vision for Attentiveness Monitoring

12. Conclusion :

SentinelVision AI is a real-time, browser-based attentiveness monitoring system that detects prolonged eye closure, hidden faces, and person absence using an ordinary webcam. By calculating the Eye Aspect Ratio and applying configurable time thresholds, it separates natural blinking from genuine alerts and avoids false alarms.

The system raises an audible alarm, records every incident with its details, lets the user adjust all key thresholds, and exports the history as CSV. Its dashboard presents all information clearly in a modern, responsive interface, and all processing stays on the user's device to protect privacy.

The prototype provides a strong foundation for future improvements such as yawn detection, analytics, notifications, and multi-person monitoring. SentinelVision AI shows how low-cost, privacy-friendly computer vision can improve safety and attentiveness in everyday settings.

KEY POINTS

- Accurate, private, configurable and easy to use on any webcam.

SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

13. References and Acknowledgement :

The SentinelVision AI project was developed based on the practical need for an affordable and privacy-respecting way to monitor attentiveness. The design and implementation were guided by the following references and sources of inspiration:

1. Eye Aspect Ratio Research: The EAR method for blink and eye-closure detection from facial landmarks, described by Soukupova and Cech in "Real-Time Eye Blink Detection Using Facial Landmarks" (2016).
2. Facial Landmark Detection: Browser-compatible face landmark models, such as MediaPipe Face Mesh, used to locate eye and face points.
3. MediaDevices API Documentation: MDN Web Docs reference for getUserMedia() and enumerateDevices().
4. Web Audio API Documentation: MDN Web Docs reference for generating and controlling alarm sound in the browser.
5. Web Development Technologies: HTML, CSS, and JavaScript concepts used to build the dashboard and interactive features.
6. Driver and Attentiveness Monitoring Systems: General working principles of drowsiness and presence monitoring systems.
7. Project Development and Testing: Practical coding, interface design, testing, and tuning of thresholds to reduce false alarms.

We thank our faculty and the Department of Computer Science and Engineering, Pragati Engineering College (A), for their guidance and support.

KEY POINTS

- Core method: Eye Aspect Ratio from facial landmarks (Soukupova and Cech, 2016).

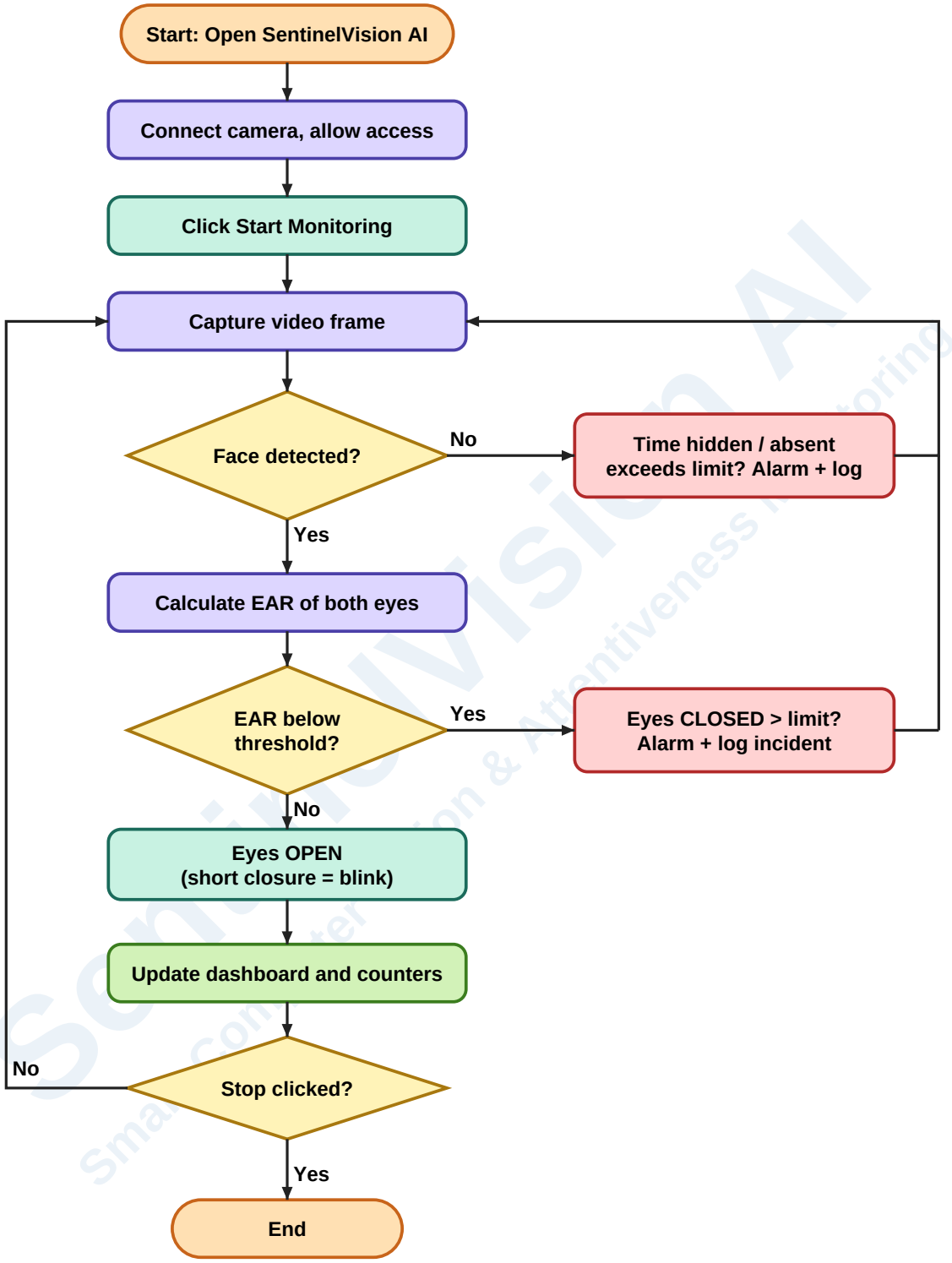
SentinelVision AI
Smart Computer Vision & Attentiveness Monitoring

14. Algorithm of SentinelVision AI :

1. Start the application and open the SentinelVision AI dashboard.
2. Connect the camera, select the required device, and allow browser permission.
3. Load the saved settings (or the default settings).
4. When the user clicks Start Monitoring, set the status to Monitoring and start the monitoring timer.
5. Capture the next video frame.
6. Run face and landmark detection on the frame.
7. If no face is detected, increase the face-hidden / person-absent timer; otherwise reset it.
8. If the timer exceeds its threshold, trigger the alarm, update the counter, and record the incident.
9. If a face is detected, calculate the EAR of both eyes and take the average.
10. If EAR is below the sensitivity threshold, mark the eyes CLOSED and increase the eyes-closed timer.
11. If EAR is above the threshold, mark the eyes OPEN. If the closure was shorter than the limit, count it as a natural blink and reset the timer.
12. If the eyes-closed timer exceeds the duration threshold, trigger the alarm, update the counter, and record the incident.
13. If normal conditions resume and Auto-Silence is on, stop the alarm and complete the incident record.
14. Update the dashboard indicators.
15. Repeat from step 5 until the user clicks Stop Monitoring.
16. Stop the camera processing, set the status to Standby, and end.

KEY POINTS

- Repeat per frame: detect face, calculate EAR, check timers, alarm, log, update dashboard.

SentinelVision AI
Smart Computer Vision for Attentiveness Monitoring

15. Flowchart :